

COMMUNITY SERVICE PROJECT REPORT

Abstract

Title

Immersive AR and VR Farming Assistant for Community-Based Smart Agriculture

Aim

The aim of this community service project is to design and develop an affordable, accessible, and technology-driven farming assistant that supports farmers in identifying crop diseases, understanding treatment methods, and improving agricultural productivity through the integration of Artificial Intelligence, Augmented Reality (AR), and Virtual Reality (VR) technologies.

The project focuses on empowering rural farming communities by providing modern agricultural guidance using smartphones without requiring costly devices or advanced technical knowledge.

Objectives

The major objectives of the project are:

To assist farmers in early identification of plant diseases using image-based detection techniques.

To provide real-time Augmented Reality visualization that highlights infected areas on plant leaves.

To offer practical solutions such as organic and chemical treatment recommendations.

To create an interactive Virtual Reality learning environment for agricultural education and training.

To enable farmers to access information without installing mobile applications through a browser-based system.

To support offline functionality for rural areas with poor internet connectivity.

To provide voice guidance in regional languages for easy understanding.

To promote awareness about sustainable farming practices and responsible pesticide usage.

To bridge the gap between agricultural experts and rural farmers using digital technology.

To enhance community participation and technology adoption in agriculture.

Tools and Technologies Used

1. Frontend Technologies

HTML5 – Structure of web application

CSS3 – Styling and user interface design

JavaScript – Interaction and functionality development

2. Augmented Reality and Virtual Reality

A-Frame

Used to create 3D environments and immersive VR scenes.

AR.js

Enables Augmented Reality directly in mobile browsers without application installation.

3. Artificial Intelligence Technologies

TensorFlow Lite – Lightweight machine learning framework

Convolutional Neural Network (CNN) – Image classification model for disease detection

4. Dataset

PlantVillage Dataset used for training crop disease detection models.

5. Backend and Storage

Firebase (optional cloud storage and database)

Offline JSON files for rural offline access

6. Hardware Requirements

Smartphone with camera

Internet browser

Low-cost VR cardboard headset

Overview of the Project

Agriculture is the backbone of rural communities, yet farmers frequently face challenges such as crop diseases, lack of expert consultation, delayed diagnosis, and limited access to agricultural education. Traditional disease identification relies heavily on agricultural specialists, which may not be easily accessible in remote villages.

This community service project introduces an Immersive AR + VR Farming Assistant, a web-based digital solution aimed at assisting farmers directly in the field.

The system works by allowing farmers to open a web link on their smartphone and scan crop leaves using the device camera. Artificial Intelligence algorithms analyze the leaf image and detect possible diseases. Once detected, Augmented Reality overlays visually highlight the infected region on the plant itself using color-coded indicators such as red (severe), yellow (moderate), and green (healthy).

Farmers can then view detailed information including:

Disease name

Causes

Preventive measures

Recommended treatments

Organic and chemical remedies

Voice-based instructions in local languages

A unique feature of the project is the AR Simulation Module, where users can visualize how diseases spread if untreated and observe crop recovery after applying recommended treatments. This visual learning approach improves understanding compared to traditional text-based information.

Additionally, a Virtual Reality Training Mode is developed for community learning programs. Farmers, students, and agricultural volunteers can enter a virtual farm environment where they practice identifying diseases and selecting treatments. Incorrect decisions demonstrate crop damage, while correct actions show improved crop health.

The system also supports offline functionality by storing essential information locally, ensuring usability even in areas with limited internet connectivity.

The project serves as a community service initiative by combining education, technology, and agriculture to improve rural livelihoods.

Community Service Activities Performed

Interaction with local farmers to understand agricultural challenges.

Awareness sessions about crop disease identification.

Demonstration of AR-based plant scanning.

Training farmers on using smartphones for agricultural assistance.

Promotion of sustainable and scientific farming practices.

Providing digital agricultural education through VR simulations.

Outcome of the Project

The project achieved the following outcomes:

Farmers gained awareness about early disease detection.

Improved understanding of crop management practices.

Reduced dependency on agricultural experts for basic diagnosis.

Increased confidence among farmers in using digital tools.

Enhanced learning experience through visual AR and VR technologies.

Promotion of sustainable farming and reduced crop losses.

Benefits to the Community

Easy access to agricultural knowledge.

Low-cost technological support for farmers.

Improved crop productivity awareness.

Digital literacy development in rural communities.

Practical training without risk to real crops.

Conclusion

The Immersive AR + VR Farming Assistant demonstrates how modern technologies can be effectively used as a community service initiative to address real-world agricultural challenges.

By integrating Artificial Intelligence, Augmented Reality, and Virtual Reality into a simple web-based platform, the project empowers farmers with accessible, understandable, and practical agricultural guidance.

The project not only improves farming awareness but also promotes digital inclusion and sustainable agriculture within rural communities. With further development and government or institutional support, this solution can be expanded to serve larger farming populations, contributing to improved agricultural productivity and rural development.

8-Week Action Plan – Immersive AR and VR Farming Assistant

Week 1 — Planning & Requirement Analysis

Goals

Finalize project idea and features

Define problem statement

Select crops & diseases (3–4 only)

Decide tools and technologies

Tasks

Study existing farming apps

Prepare system architecture

Install development tools

Create GitHub/project folder

Week 2 — Dataset Collection & Preparation

Goals

Prepare AI training data

Tasks

Download leaf images from PlantVillage

Clean dataset

Label disease categories

Resize images

Split training/testing data

Week 3 — AI Model Development

Goals

Build disease detection model

Tasks

Train CNN model

Test accuracy

Optimize model size

Convert model → TensorFlow Lite

Week 4 — AR Diagnosis Module

Goals

Enable plant scanning in browser

Tools

A-Frame

AR.js

Tasks

Camera access via browser

Detect leaf image

Add AR overlay highlighting disease

Show disease info panel

Week 5 — AR Simulation Feature

Goals

Create innovation feature

Tasks

Design 3D plant model

Animate:

Disease spread

Treatment recovery

Add “Simulate Treatment” button

Week 6 — VR Training Mode

Goals

Build immersive learning environment

Tasks

Create virtual farm scene

Add diseased plants

User selects treatment

Show correct/wrong result

Week 7 — Offline & Smart Features

Goals

Make system farmer-friendly

Tasks

Offline caching (Service Worker)

Local JSON remedies database

Voice/audio guidance

Performance optimization

Week 8 — Testing, Deployment & Documentation

Goals

Final project completion

Tasks

Functional testing

Accuracy testing

Deploy website

Prepare:

Report

PPT

Demo video